

MON TUE
WED THU
FRI SAT
SUNDAY

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17 18 19 20 21 22 23 24
25 26 27 28 29 30 31 -

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Positional mapping:-

$$\pi: B \rightarrow G$$

$$G = \{(x, y)\}$$

Zone Assigning:-

$$\tau: B \rightarrow Z$$

$$Z = \{z_1, z_2, \dots, z_m\}$$

zone or superblock

C: core cells

A: anchor cells

Objective func:-

$$U(\pi, \tau) = w_1 \cdot U_{\text{proximity}}(\pi)$$

$$+ w_2 \cdot U_{\text{adjacency}}(\pi, \tau)$$

$$+ w_3 \cdot U_{\text{travel}}(\pi, \tau)$$

$$+ w_4 \cdot U_{\text{composition}}(\tau)$$



$w_i \geq 0$ and $\sum w_i = 1$

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Constraints

① Hard

Unique Position Assigned

$$\pi(b_i) \neq \pi(b_j) \quad \forall i \neq j$$

π should be injective

Valid Grid Positions

$$\pi(b) \notin C$$

blocks should not be mapped
to core positions

Connected superblocks/zones

let B^z is set of blocks in zone z

then $\forall z \quad \mathcal{G}\{\pi(b) \mid b \in B^z\}$

should be connected graph 0%

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② Soft Constraints:-

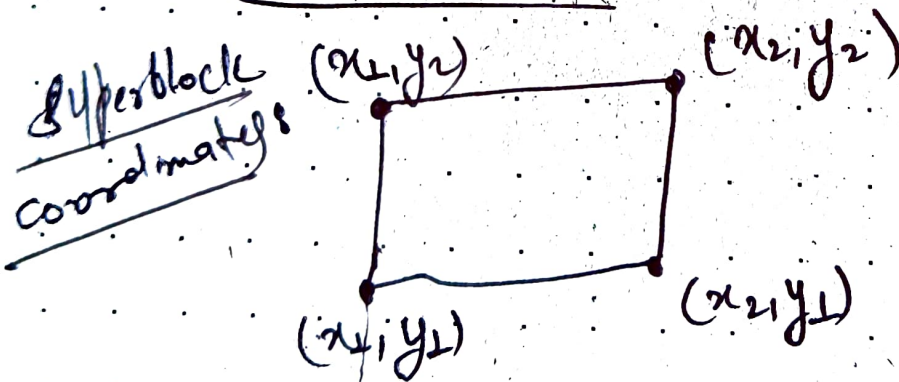
Zone size Bound

there is min and max size of zone give in excel.

Diversity constraint :-

no dept. should exceed more than 70% of a superblock

Aspect ratio :-



$$\text{height} = y_2 - y_1 + 1$$

$$\text{width} = x_2 - x_1 + 1$$



Aspect ratio

Constraint:-

$$\frac{1}{\gamma_{\max}} \leq \frac{\text{width}}{\text{height}} \leq \gamma_{\max}$$

Greedy Algo for Positioning

Input $\rightarrow B, F = \{W, H, C, A\}$

Output $\rightarrow \pi$

Algo:

① $U \leftarrow \phi$ (set of used cells)
 $\pi \leftarrow$ empty mapping

② Sorting Blocks by priority

$B_1 \leftarrow$ destination (US, destination)

$B_2 \leftarrow$ speciality

$B_3 \leftarrow$ support

$B_4 \leftarrow$ ME

$B_5 \leftarrow$ WE



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③ for each Priority Group B_i in order

- Group by department
- for each department
 - sort by travel group
 - for each Block b
 - we find best cell g^*

$$g \in \underbrace{(G \setminus C)}_{\setminus v}$$

$$g^* = \arg \max \{ \text{score}(g) \}$$

here $\text{score}(g) =$

$$\begin{aligned}
 & w_{\text{prox}} \cdot \text{Proximity Score}(b, g) \\
 & + w_{\text{dept}} \cdot \text{Dept. Cluster Score}(b, g, \pi) \\
 & + w_{\text{neighbour}} \cdot \text{Neighbourhood Score}(g, v)
 \end{aligned}$$



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$$\pi(b) \leftarrow g^*$$

$$U \leftarrow U \cup \{g^*\}$$

④ return π

T.C. = $O(n \cdot w \cdot h)$

Spatial Clustering Algorithm

For creating super blocks

Disjoint Set Union (Union Find)

① Initialise:

$\text{parent}(b) \leftarrow b$ (each Block is own parent)

② Connection predicate:-

$\text{Connect}(b_1, b_2) \Leftrightarrow (\text{adj}(\pi(b_1), \pi(b_2)) = 1)$

$\neg (d(b_1) = d(b_2) \vee s(b_1) = s(b_2))$

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- means we connect b_1, b_2 if their adjacency $= 1$ and either their dept. is same or space type is same.

③ Build connection components :-

for each $(b_1, b_2) \in B \times B$

if $(\text{connect}(b_1, b_2))$:

Union (b_1, b_2)

↳ make their parent same

④ clusters :-

for each b :

$\text{root} \leftarrow \text{find}(b)$ ← parent finder

$\text{cluster}[\text{root}] \leftarrow \text{cluster}[\text{root}] \cup \{b\}$



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⑤ Refine clusters

$Z \leftarrow \phi$

~~Zone~~ (zone)

$z_id \leftarrow 1$

for each cluster C in cluster.values():

if $|C| > n_{max}$: (max zone size)

sub $\leftarrow$ split by Attribute (C , "dept")

for each $S \in sub$:

$Z \leftarrow Z \cup \{Zone(z_id, S)\}$

$z_id \leftarrow z_id + 1$

else:

$Z \leftarrow Z \cup \{Zone(z_id, C)\}$

$z_id \leftarrow z_id + 1$

⑥ Return Z .

F.T.C. = $O(n)$



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(expand zone cells).

Zone Boundary Expansion :-

If $(x, y) \notin C$ has adjacency in
(superblock z)

$$\text{then } G^z \leftarrow G^z \cup \{(x, y)\}$$

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For getting Target composition of SuperBlocks

let Target $\tau = \{ \tau_{ME}, \tau_{WE},$
 $\tau_{support}, \tau_{speciality},$
 $\tau_{us} \}$

$$\sum \tau_s = 1$$

① ideal counts

for each spacemix $\hat{n}_s \leftarrow \tau_s \cdot N$ → total blocks

round off $\hat{n}_s$ to integer

$$\text{deficit} = N - \sum n_s$$



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for $i \leftarrow 1$ to deficit:

~~$n_{s_i} + 1$~~ $n_{s_i} + 1 = 1$;

(s_i is i -th spacemix
according to sorted
priority order)

Return $\{n_s\}_s \in S$.

Score of
Composition Matching:-

$$C_h(z, \tau) = \frac{1 - (\sum |\hat{n}_s - n_s|)}{N}$$

target ↑
actual ↑



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Overall Weighted Score: →

$$\begin{aligned} \text{Score}(z, \tau) = & w_1 \cdot \text{Density}(z, \tau) \\ & + w_2 \cdot \text{Adjacency}(z) \\ & + w_3 \cdot \text{Travel}(z) \\ & + w_4 \cdot \text{Diversity}(z) \end{aligned}$$

$$\sum w_i = 1$$